



ECDS



DATAQUEST 3.0

DQBH

Title

Next-Generation Activity Management Platform for Industrial Equipment Services

Background

Design and build a centralized, extensible full-stack activity management platform for industrial equipment servicing and maintenance across multi-site operations, ensuring that it satisfies both:

1. Full-Stack Application Development — Develop a complete and scalable web application with frontend, backend, database, authentication, authorization, APIs, business logic, workflow management, dashboards, notifications, and audit history.
2. Extensible Technology Architecture — Design the platform so that mobile applications, AI/ML, IoT, real-time communication, analytics, maps, blockchain, and cloud services can be integrated as additional capabilities.

Consider an organization that manages the servicing and maintenance of industrial equipment across multiple customer sites. Currently, service requests, technician schedules, spare parts inventories, and completion verifications are handled through disconnected tools, spreadsheets, and messaging apps. If Machine M-104 at Site A requires urgent maintenance, operations teams must manually verify technician and part availability, handle sudden technician dropouts, and manage uncoordinated delays. This fragmented process prevents real-time visibility, creates resource bottlenecks, and compromises maintenance traceability. The organization requires a single full-stack digital platform to manage the complete equipment service lifecycle.

Problem

Industrial equipment service operations often suffer from:

- Service requests being recorded and communicated through multiple disconnected systems.

- Difficulty identifying the right technician, tools, spare parts, or other resources for a service request.
- Limited real-time visibility into technician assignments, machine status, service progress, and pending requests.
- Delays or resource conflicts being identified only after they have already affected the service schedule.
- Manual collection and verification of service reports, photographs, documents, and completion information.

These issues lead to delayed maintenance, inefficient resource utilization, duplicated work, poor visibility, and difficulty reconstructing machine service history. The proposed platform provides a centralized system where machines, service requests, technicians, resources, documents, approvals, and completion records are managed through one connected application.

Challenge

Design and develop an activity management platform to coordinate equipment servicing and operational workflows across multiple locations, ensuring that it satisfies both:

1. Core full-stack workflow and resource management
2. Extensible architecture for advanced operational capabilities

1. Operational Scope:

- Creation, tracking, and lifecycle management of equipment service requests.
- Allocation and coordination of field technicians and operational resources.
- Conflict identification, status updates, and verification records.

2. Workflow Scenario:

Scenario:

Managing an unscheduled maintenance request across field sites. The platform should:

- Route requests to suitable personnel based on availability and location.
- Enable status updates, task logging, and completion verification.
- Provide centralized operational visibility across pending and active tasks.

Core Concepts

The core workflow begins when a customer or authorized operations user creates a service request:

- The system validates machine eligibility, priority, site location, required skills, and spare parts availability before routing for approval.

- Once approved, the platform assigns qualified technicians and reserves resources, dynamically notifying field personnel and updating the timeline.
- If a technician drops out, a part is unavailable, or work exceeds SLA, the system raises an exception flag, triggers alerts, and supports instant reassignment.

Wishing all participating teams the very best of innovation and success at DataQuest 3.0!